

University of Dhaka
Department of Biomedical Physics and Technology
M.S. in Biomedical Engineering
Session 2024-25
BMPT 514: Biomedical Instrumentation Laboratory
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Experiment No. 01

Name of the experiment: Study the different stages of an ECG acquisition circuit and the measurement of limb lead ECG.

Objectives:

- To get acquainted with the circuit diagram and data acquisition process in the ECG trainer kit.
- To observe the trace of the six frontal plane ECG.

Theory:

Electrocardiography (ECG) is the process of visualizing the heart's electrical activity over time. These electrical activities result from the depolarization of the myocardium in a certain pattern. The electrical activities are monitored using surface electrodes over the body.

As we know, the heart has a certain rhythm over a course of time in which it contracts and expands, pumping blood out to the body. This periodic rhythm is caused by depolarization of some specific points on the myocardium, namely the Sino-Atrial Node (SA node), the Atrio-Ventricular Node (AV node), and the His-Purkinje Fibers. These depolarizations create electrical impulses. As the human body is a volume conductor, the effect of any electrical impulse inside the body can be detected from the surface.

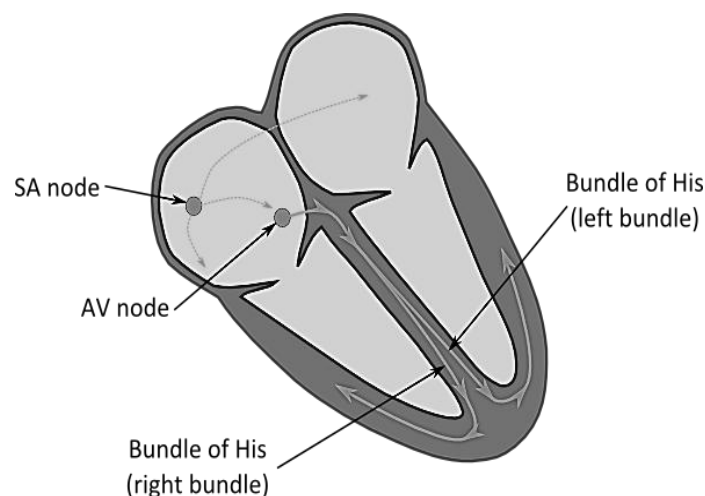


Figure 01: Nodes in the myocardium

The electrical activity of the heart can be modelled as the movement of an electrical dipole. A dipole consists of two opposite charges. A cardiac vector is the line joining the two charges. Although the heart does not consist of two opposite charges, the electric field created due to the depolarization of the heart resembles that of an electric dipole. To describe a vector by its length in three dimensions, three directions are considered, which are 60° apart. These directions form a triangle known as Einthoven's triangle. The direction from one point to another is known as a Lead, hence having three leads.

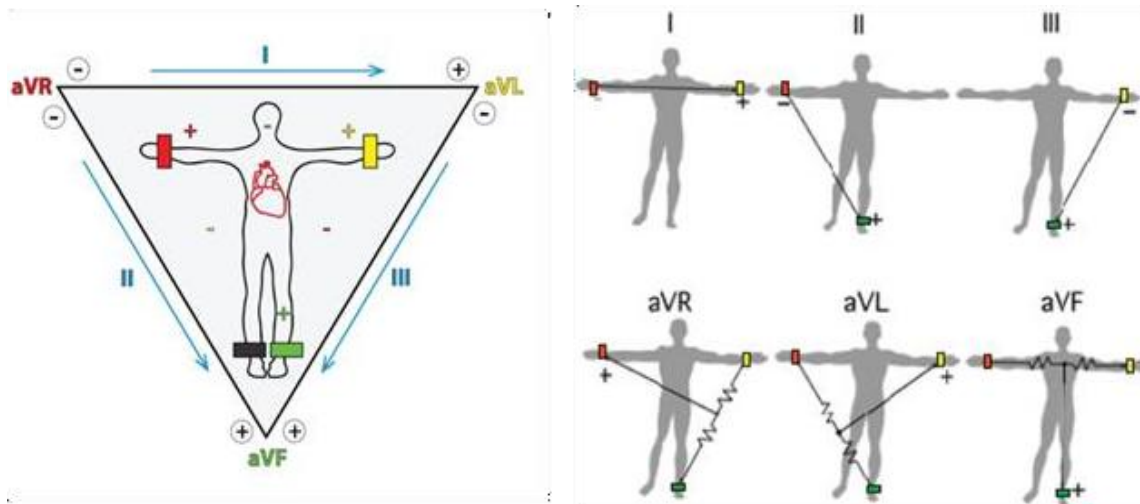


Figure 02: Configuration of different frontal leads

The body and the heart are three-dimensional, so the electrical signals recorded from the skin will vary depending on the position of the electrodes. For diagnostic purposes, some standardization of electrode positioning is needed. This is done by placing the electrodes on the limbs. The three points of the Einthoven's triangle are the three limbs where we put the electrodes. The three leads are thus considered as,

Lead I: from Right Arm (-ve) to Left Arm (+ve)

Lead II: from Right Arm (-ve) to Left Leg (+ve)

Lead III: from Left Arm (-ve) to Left Leg (+ve)

Apart from the three main leads, there are three other leads in the frontal plane named Augmented leads. These are denoted as aVR, aVL, and aVF. They are derived from the same three electrodes as leads I, II, and III, but they use Goldberger's central terminal as their negative pole. Goldberger's central terminal is a combination of inputs from two limb electrodes, with a different combination for each augmented lead. It is referred to immediately below as "the negative pole."

Lead aVR: $RA - (LA + LL)/2$

Lead aVL: $LA - (LL + RA)/2$

Lead aVF: $LL - (RA + LA)/2$

As the lower half of the body can be considered equipotential, the bottom point in the Einthoven's triangle can be considered as any point in the lower thorax. For simplicity, we use the left leg for this point.

To reduce the effect of noise, the average of the signals is fed back to the body via an instrumentation amplifier. This cancels out the common-mode noise interference. As the left hand, right hand, and left leg are used to record the various leads, this average signal is fed back to the body via the right leg. Hence, this configuration is known as a right-leg-driven circuit.

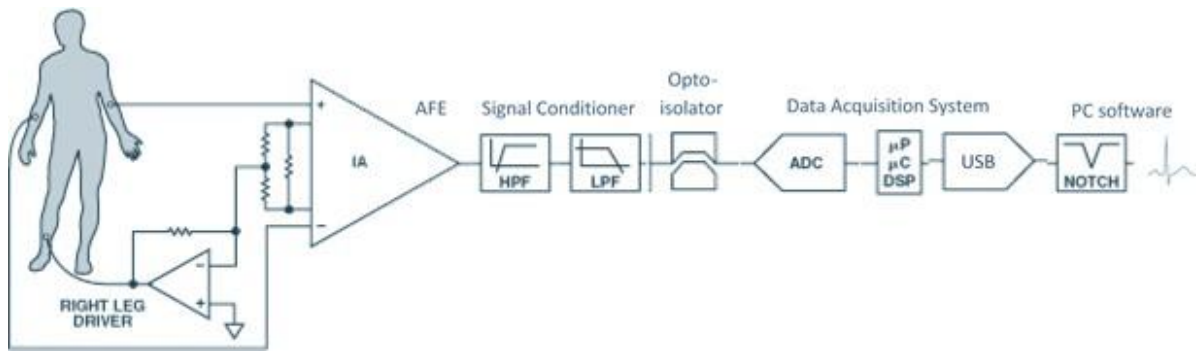


Figure 03: Block diagram for ECG trainer module

Equipment:

- ECG trainer module
- Computer / Oscilloscope
- Multimeter

Setup:

A full layout of the experimental setup is presented in **Figure 04**.

The connection setup for various lead configurations is shown in **Figure 05**.

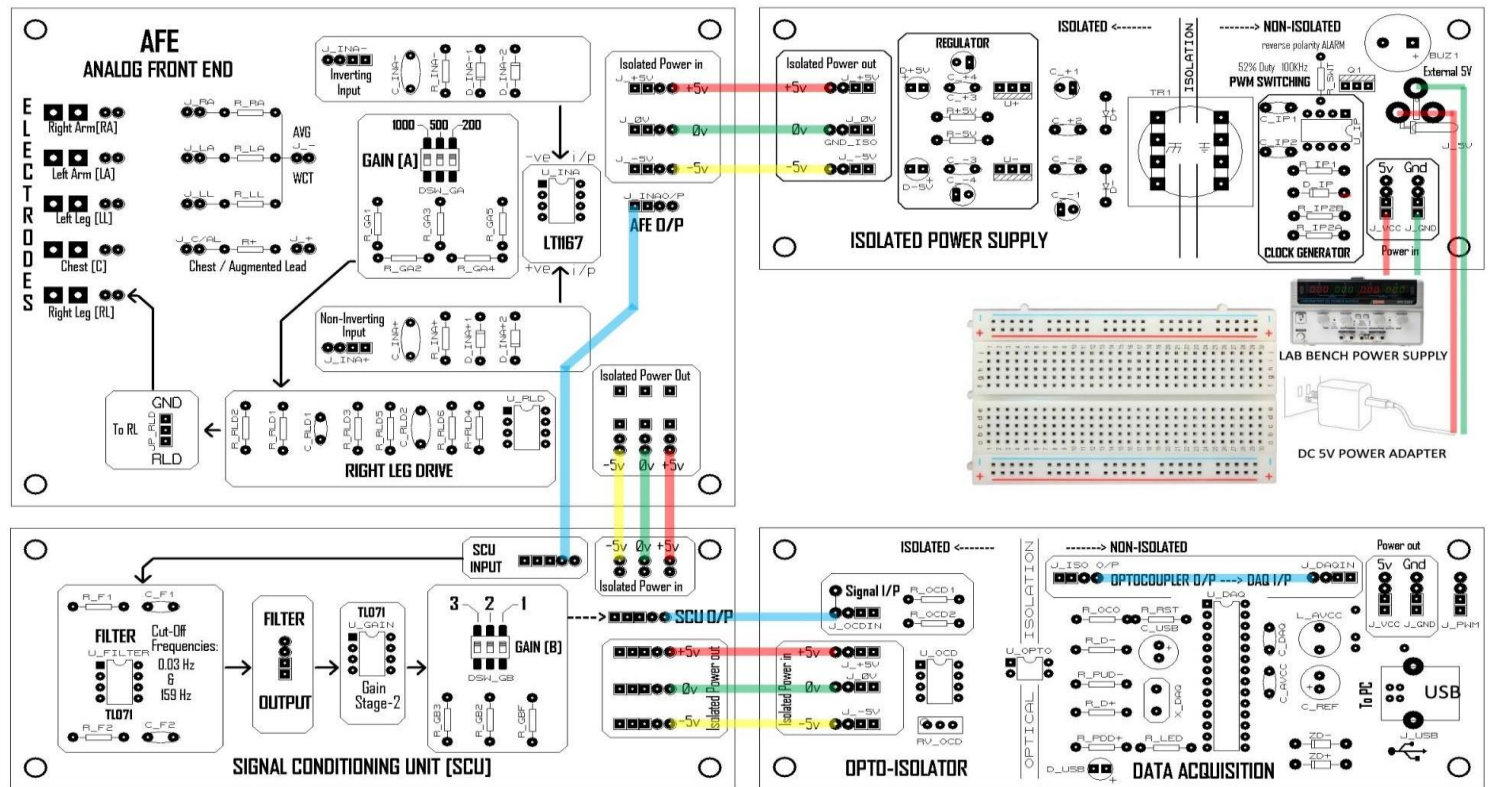
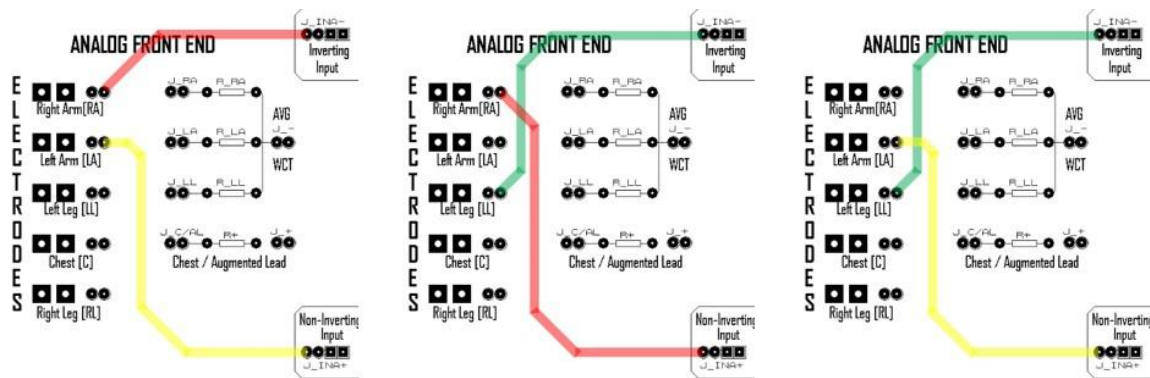


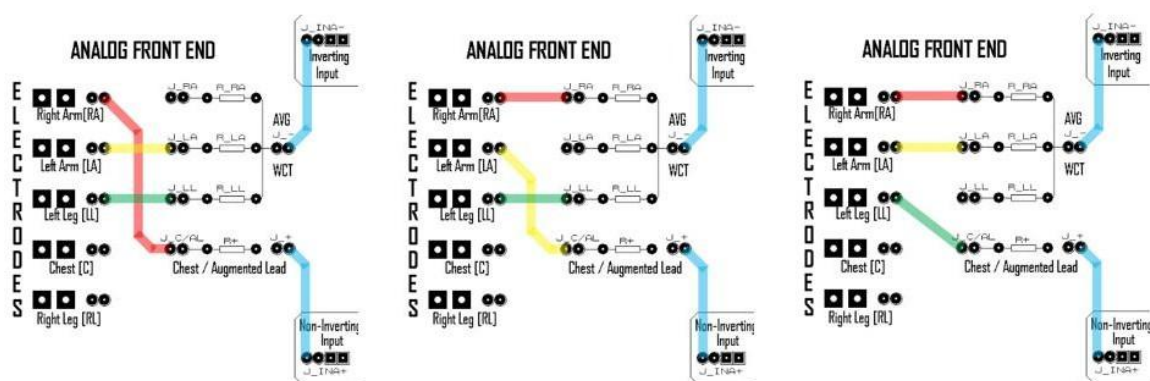
Figure 04: Circuit setup of ECG trainer kit.



Lead – I configuration

Lead – II configuration

Lead – III configuration.



Lead aVR configuration

, Lead aVL configuration, Lead aVF configuration.

Figure 05: Connection for various leads in the ECG Trainer module**Results:**

- Take the snapshot of the ECG leads from the computer of the oscilloscope for the various leads
- Study the change in output trace due to the variation of the input gain
- Study the variation of output trace due to the effect of the Right Leg Driven circuit and the Ground Driven circuit.

Discussion:

Discuss the problems, findings, shortcomings, outcome discrepancies of the experiment with a theoretical explanation.

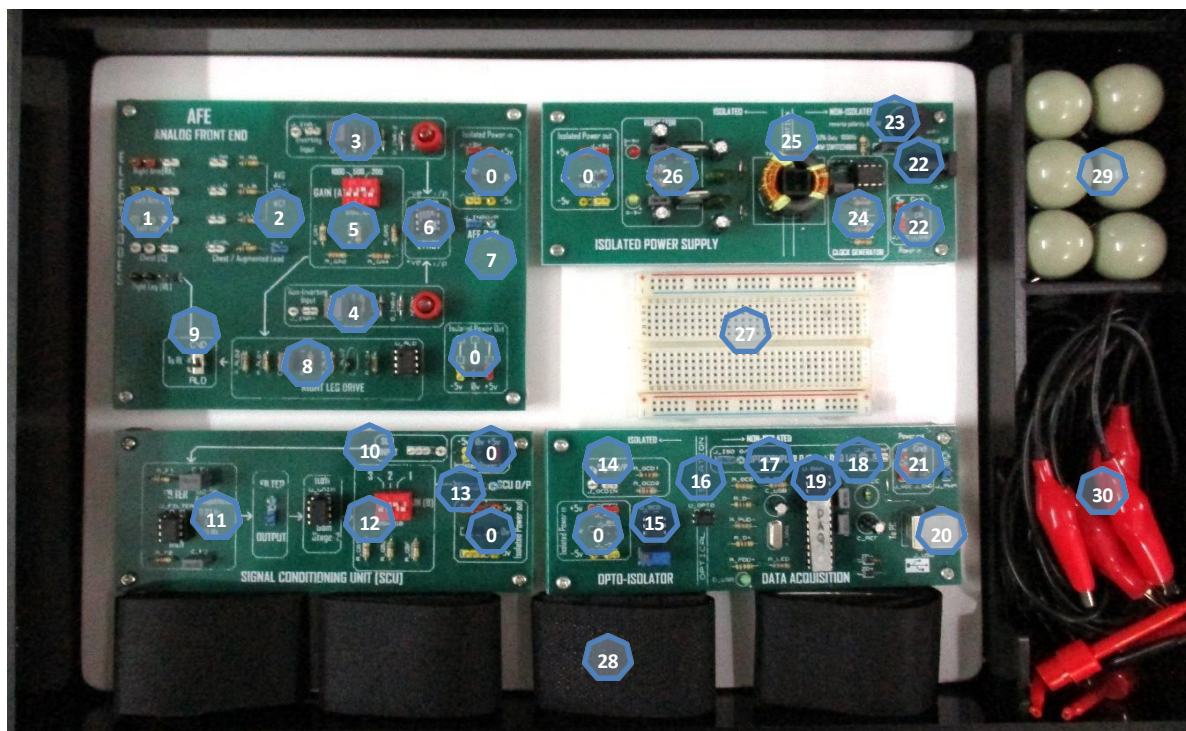


Figure 06: Overview of ECG trainer module

Experiment No. 02

Study the different stages of an EMG acquisition circuit and the measurement of EMG.

Objective:

- To get acquainted with the circuit diagram and data acquisition process in the EMG trainer kit
- To observe trace of EMG.

Theory:

EMG stands for Electromyography – meaning visualization of the electrical signals of muscles. A mass of muscle that we feel at different places in our body consists of many muscle fibers. At rest each muscle fiber has a charge separation across its covering membrane (outer surface positive with respect to the inside of the fiber), giving rise to a polarized state. This is reversed or depolarized in segments of the fiber when an action is initiated, which after a relaxation period, automatically reverts back to the resting polarized state. This contributes to an action potential which travels along the fiber. At any instant of time this charge separation contributes to an electric dipole vector. Similar dipole vectors are produced in all other muscle fibers in the group. A muscle is usually controlled by nerve signals (nerve action potentials) sent from the brain. When nerve action potentials reach the muscle, muscle action potentials are initiated through the neuromuscular junctions, which give rise to EMG. Such an EMG is also called voluntary EMG since it is the result of a voluntary action of the person.

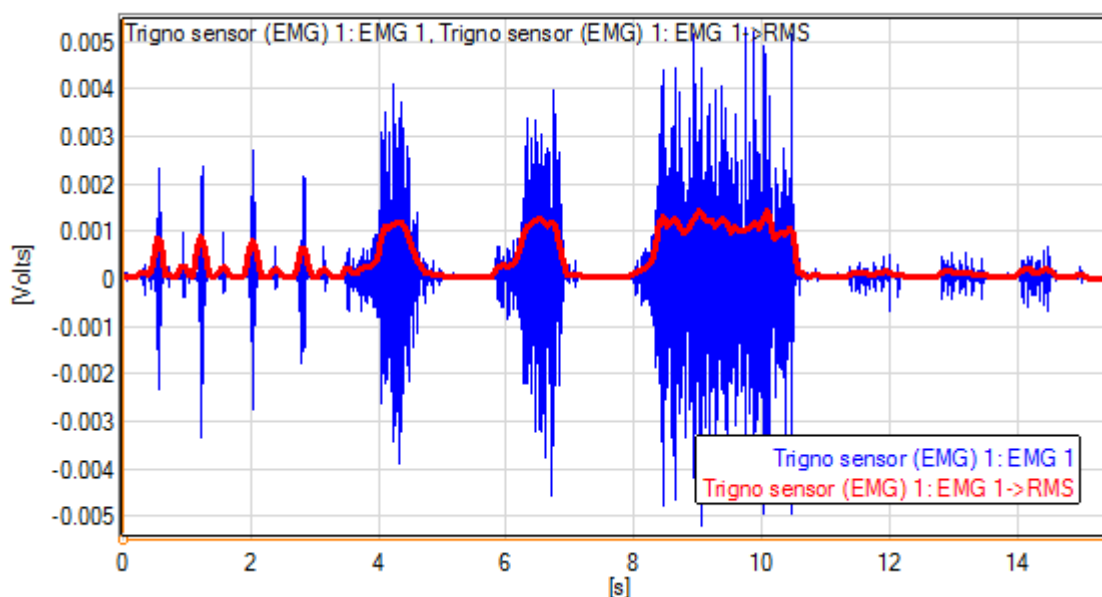


Figure 01: Typical EMG signal for voluntary muscle contraction

The body is a volume conductor, consisting of electrolytes, having both positive and negative free ions. Therefore, within the body, the dipole vectors give rise to an external current which changes with time. When a current flows through a resistor, a potential drop occurs across it. Similarly, due to the external current distribution in the volume conductor, potential differences are produced between any two points. This potential difference can be picked up using two

electrodes placed at two suitable points outside, which, when plotted against time, gives the well-known EMG as shown in **Figure 01**.

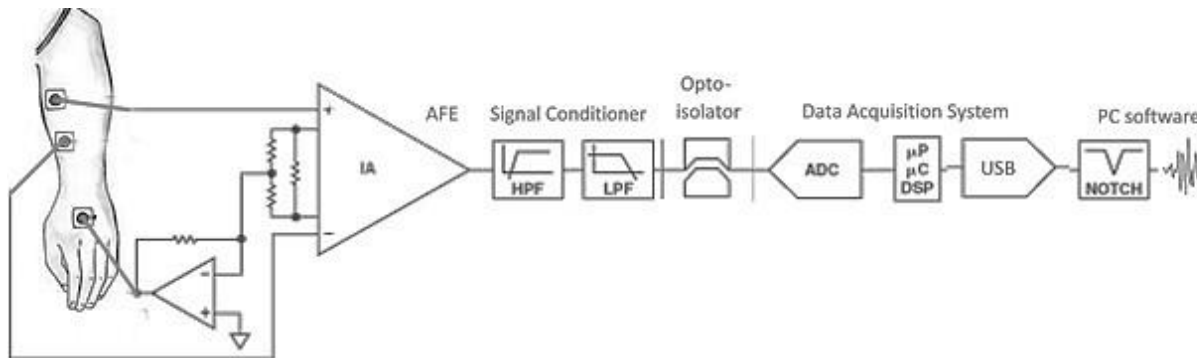


Figure 02: Block diagram for EMG trainer module

This experiment represents the evaluation of the full EMG circuit. Different modules will be connected together to acquire the EMG signal from the body and to reconstruct it in a PC software. In this experiment, students can learn about the different electrode placements and the signal chain of EMG circuitry. Signal can be observed at different levels of processing, like Bio-amp, filter output, and gain stages. Students can also study the effect of the right leg drive circuitry and implementation of a digital filter in PC software to get rid of common noise.

Equipment:

- EMG trainer module
- Computer / Oscilloscope
- Multimeter

Setup: A full layout of the experimental setup is presented in **Figure 03**.

Results:

- Take a snapshot of the EMG leads from the computer of the oscilloscope for the various leads
- Study the change in the output trace due to the variation of the input gain
- Study the variation of the output trace due to the effect of the Right Leg Driven circuit and the Ground Driven circuit.

Discussion:

Discuss the problems, findings, shortcomings, and outcome discrepancies of the experiment with a theoretical explanation.

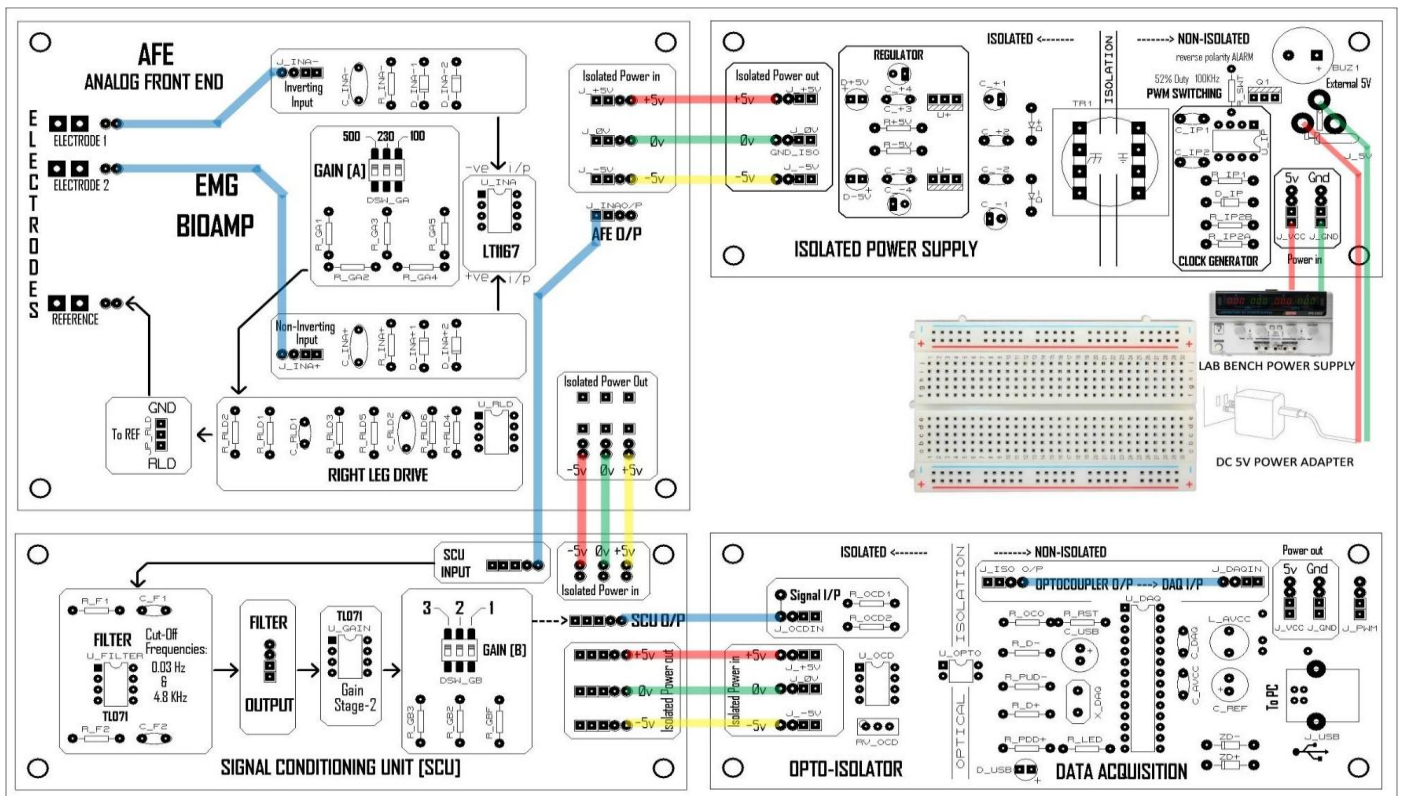


Figure 03: Circuit setup of EMG trainer kit.

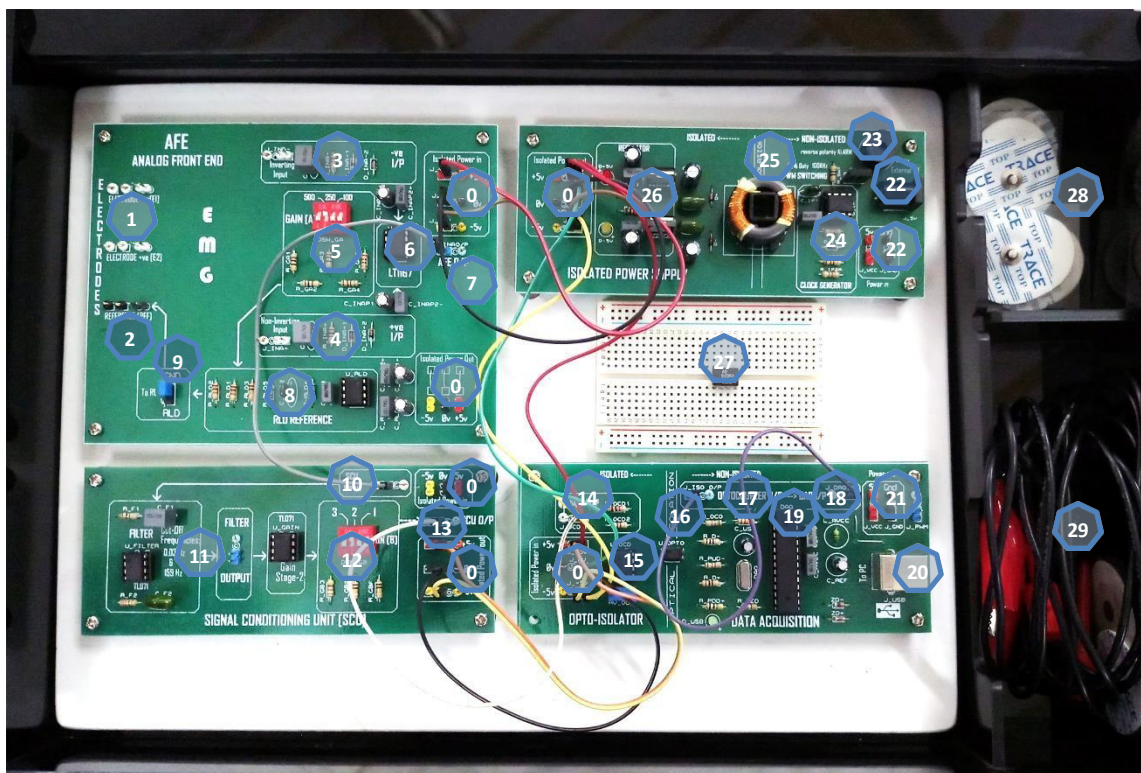


Figure 04: Overview of EMG trainer module

Experiment No. 03

Design and Analysis of Basic Filter Circuits: High-Pass, Low-Pass, and Band-Pass Filters.

Objectives:

BPF for ECG acquisition circuit.

BPF for EMG acquisition circuit.

Experiment No. 04

Study the characteristics of an opto-coupler.

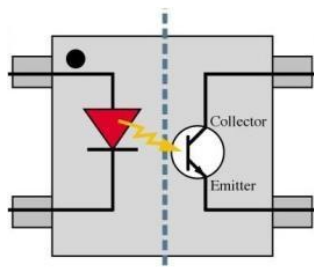


Figure 01: A basic schematic diagram of an opto-coupler IC

Objective:

- Implementation of an optical isolator using an opto-coupler circuit.
- Observe various input-output characteristics.

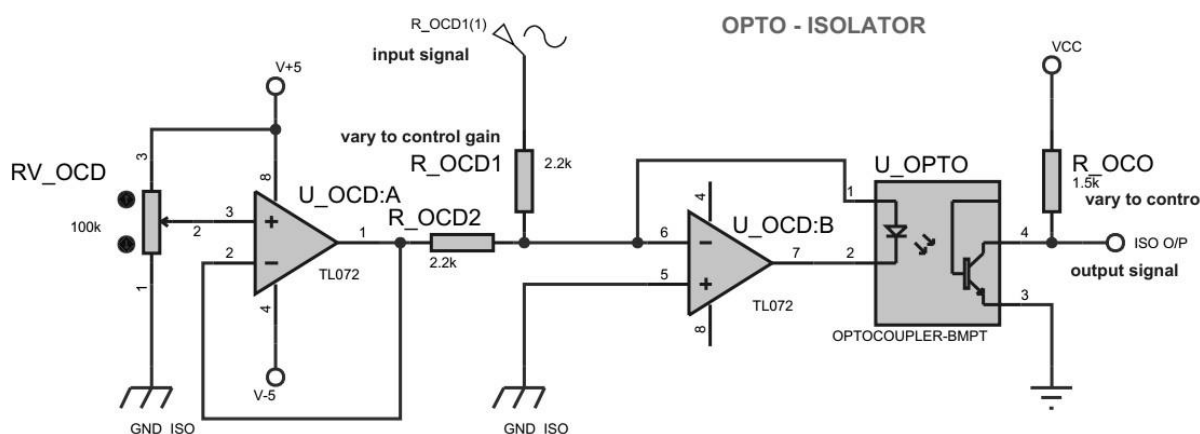


Figure 02: Circuit diagram

Theory:

An Optocoupler, also known as an Opto-isolator or Photo-coupler, is an electronic components that interconnect two electrical circuits by means of an optical interface. The basic design of an opto-coupler consists of an LED that produces infra-red light and a semiconductor photosensitive device that is used to detect this emitted infrared light. Both the LED and photosensitive device are enclosed in a light-tight body or package with metal legs for the electrical connections as shown.

An opto-coupler IC is shown in **Figure 01**. Current from the source signal passes through the input LED which emits an infra-red light whose intensity is proportional to the electrical signal. This emitted light falls upon the base of the photo-transistor, causing it to switch-ON and conduct in a similar way to a normal bipolar transistor. The base connection of the phototransistor can be left open for maximum sensitivity or connected to ground via a suitable external resistor to control the switching sensitivity making it more stable.

Equipment: (Specify the model and quantities)

- Op-Amps
- Resistors
- Variable resistors
- DC power source
- Signal Generator
- Oscilloscope

Results:

- For an input of 1V (p-p) sine wave, show the output of the opto-isolator. Vary the frequency from 50 Hz to 500Hz and show the output graphs.
- Include graph for IC vs VCE for various values of RC.

Discussion:

Discuss the problems, findings, shortcomings, outcome discrepancies of the experiment with theoretical explanation.

Experiment No. 05

Study the characteristics of the Howland circuit.

Objective:

- Design and implement Howland current sources
- Comparison between performances

Theory:

In biomedical application, it is more often desired to use a constant current source in the course of measuring bioimpedance. The human body is a bulk conductor of non-uniform constituents. Any applied current through external electrodes may result in passing through various body components of different impedances. As in such case the body is the load to the circuit supplying the current, the total current drawn could vary depending upon the impedance of the body tissue. Hence a need for a constant current source is widely appreciated.

There are various sorts of constant current sources. Among them, the Howland current source is widely used for biomedical applications. The Howland current source has two versions, the Basic Howland current source and the improved Howland current source. Both the circuits are given in the **Figure 01** and **Figure 02** below.

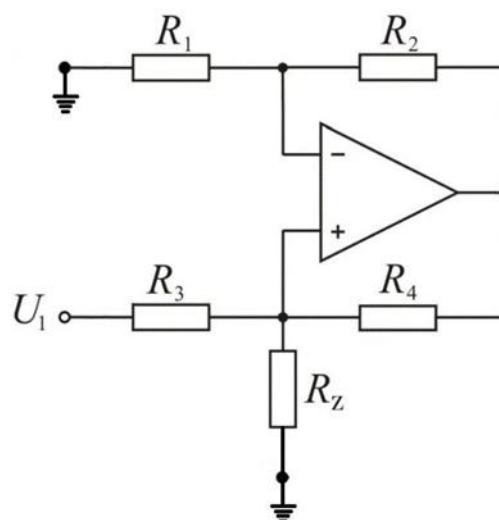


Figure 01: Basic Howland Current Source

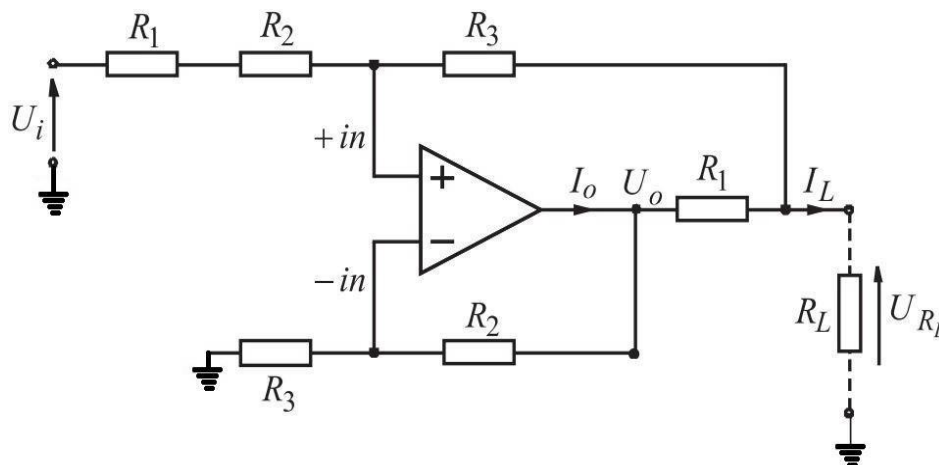


Figure 02: Improved Howland Current Source.

Equipment: (Specify the model and quantities)

- Op-Amps
- Resistors
- DC power source
- Signal Generator
- Oscilloscope

Procedure:

Assemble the two circuits given in the **Figures**. For the load impedance, take the value of R which represents a simplified model of Bioimpedance. Change the value of the load impedance by varying the value of R within the range of Bioimpedance.

For the Basic Howland current source, keep the values of $R_1 = R_2 = R_3 = R_4 = R$

For the Improved Howland current source, keep the values of R_2 to be around 10 times the value of R_1 . The values of R_2 should be equal to or close to R_3 .

Measure the load current I_L for various values of Z_L at frequencies 1kHz, 10kHz and 100kHz.

Deliverables:

1. Derive the expression for load current for the two circuit configurations.
2. Plot the Load Current vs Load Impedance for different frequencies mentioned.

Discussion:

Discuss the problems, findings, shortcomings, outcome discrepancies of the experiment with theoretical explanation.

Experiment No. 06

Contact impedance measurement using the Howland circuit as the constant current source.

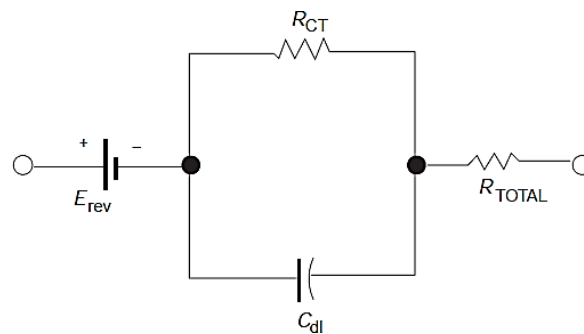


Figure 01: Contact Impedance

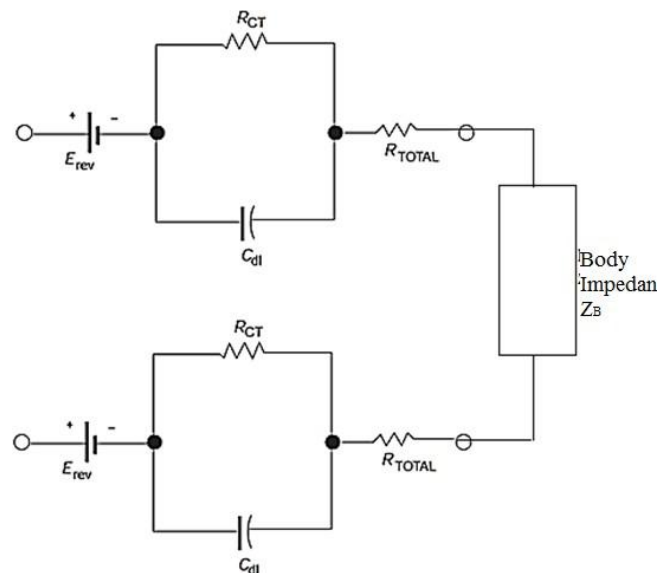


Figure 02: Total impedance of the setup

Objective:

- Establish the circuit to measure contact impedance
- To determine the frequency response of skin contact impedance.

Theory:

When applying electrical stimulation using surface electrodes, the body becomes part of an electric circuit as the current travels from one electrode to the other. The electrode-body surface interface offers frequency dependent impedance, called Contact Impedance.

In the equivalent circuit shown in **Figure 01**.

- E_{rev} is the half-cell potential
- R_{CT} and C_{dl} makes up the impedance associated with the electrode-electrolyte interface.
- R_{TOTAL} is the lump resistance associated with the resistance of electrolyte and epidermis.

The part of the body between the two electrodes constitutes the body impedance (Z_C) as shown in **Figure 02**. This impedance is dependent on supply frequency and the distance between the two electrodes.

For a certain frequency and certain electrode separation, the bulk impedance will consist of contact impedance and body impedance. If the body impedance is eliminated from the total bulk measurement, the values of the contact impedance can be obtained.

Equipment:

- Resistors
- Signal Generator
- Oscilloscope
- Electrodes
- Electrode jelly

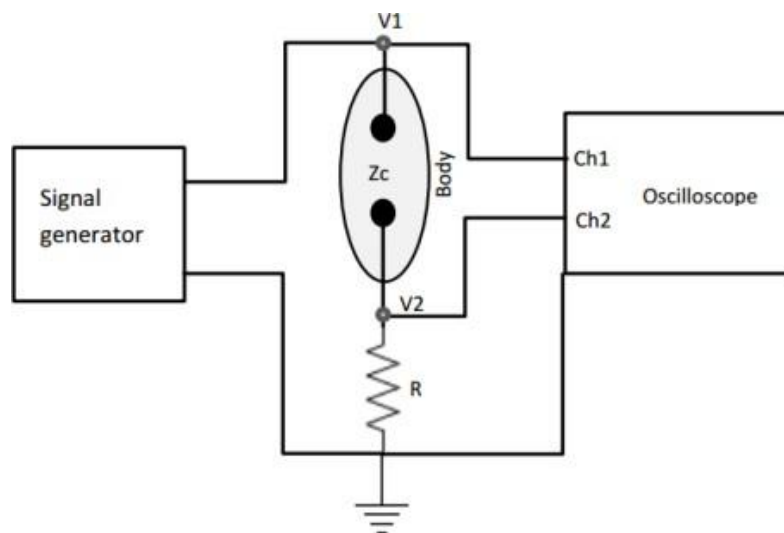


Figure 03: Circuit diagram to determine Contact Impedance

Results:

Plot the impedance vs frequency for unprepared skin, prepared skin, body impedance and contact impedance on semi-log graph paper.

Discussion:

Discuss the problems, findings, shortcomings, and outcome discrepancies of the experiment with theoretical explanation.

Experiment No. 07

To study the circuit design and breadboard implementation of a Nerve Stimulator

Objective:

- To design an astable multivibrator using 555 timer IC
- To amplify the current from the astable multivibrator using a Darlington Pair IC.

Theory:

If a current of sufficient amplitude is passed between a pair of surface electrodes then muscles will contract as high amount of currents for a short period of time causes the depolarization of the cell membrane and generates an action potential. The stimulation only occurs underneath the cathode. This is shown in the figure given below.

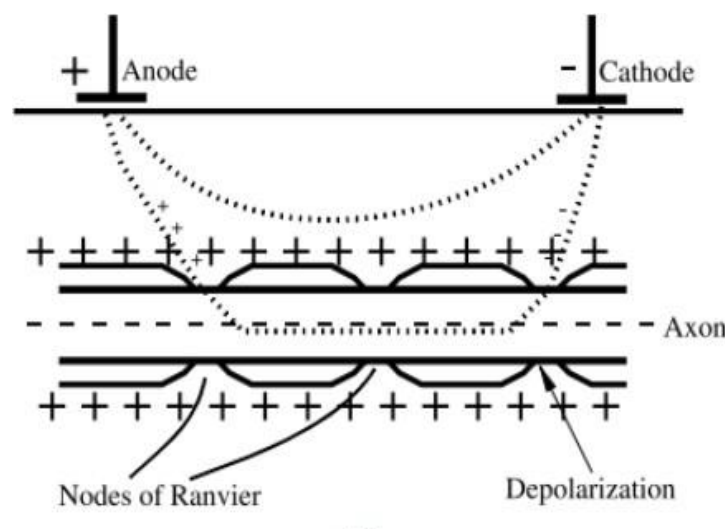
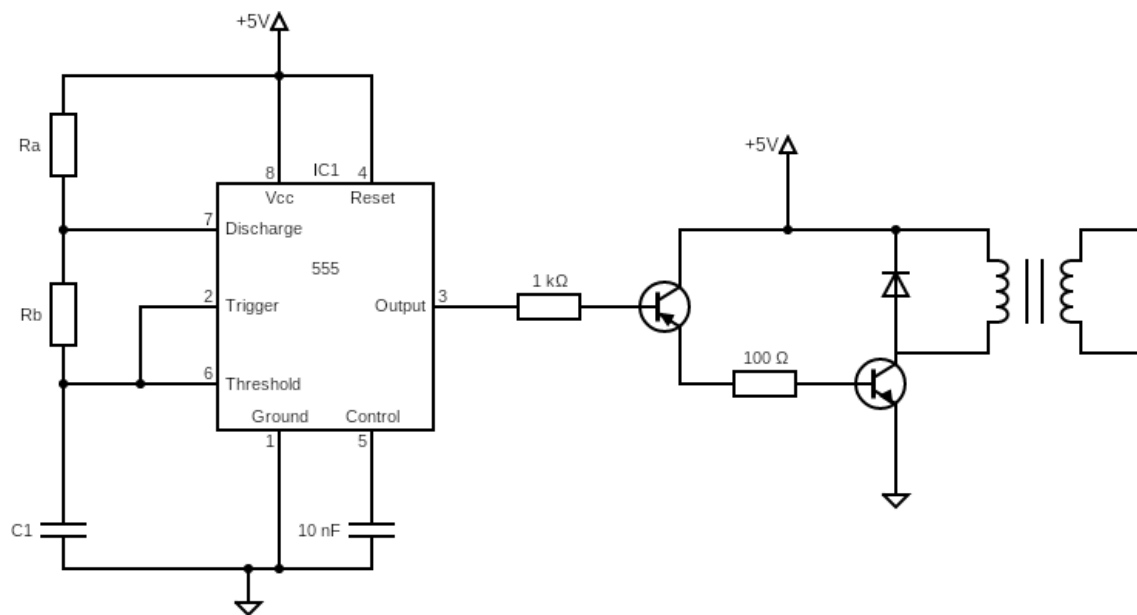


Figure 01: Stimulation under cathode reversing the membrane potential and generating action potential

An astable multivibrator will generate the required pulse train for stimulation. But the output current of the IC is very low. Hence we need to amplify the current using a Darlington Pair.

Equipment:

- Timer IC (IC 555)
- pnp and npn transistor
- Resistor
- Capacitor
- Diode
- Variable DC power supply
- Oscilloscope

Assembly:**Figure 02:** Circuit diagram

The capacitor charges through the resistors R_a and R_b which produces the high time T_{high}

$$T_{high} = 0.693(R_a + R_b) C$$

The capacitor discharges through the resistor R_b which produces the low time T_{low}

$$T_{low} = 0.693R_b C$$

The 555 timer IC has 50% duty cycle. For our application, it is required to produce a T_{high} of 1ms and T_{low} of 999ms. But such a low high time is not possible for the timer IC. So we will generate a pulse which has a high time of 999ms and a low time of 1ms and pass it through the complimentary darlington pair. The complimentary darlington pair inverts the waveform and produces our desired output pulse.

Results:

- Calculate the T_{low} and T_{high} value according to practical values of R_a and R_b .
- Find percentage of error for practical values of T_{low} or T_{high} with previous calculation.
- Observe how much current gain is achieved after complimentary Darlington section.
- Observe the waveform after the transformer.

Discussion:

Discuss the problems, findings, shortcomings, outcome discrepancies of the experiment with theoretical explanation.

Experiment No. 08

Design and characterization of an instrumentation amplifier.

Objective:

- Design an instrumentation amplifier
- Breadboard implementation
- Tune for proper gain and CMRR

Theory:

An Instrumentation amplifier is basically a differential amplifier, but with high input impedance so that it can faithfully reproduce voltage signals from any source. The idea is that there should not be significant voltage drop in the output impedance of the source, almost all the source voltage should be dropped across the input impedance of the amplifier. Instrumentation amplifiers are used in industry as well as in bioelectrical measurements.

A differential amplifier is a basic circuit to pick up and amplify a small signal in presence of a relatively large background noise. However, it is only possible if the noise is the same at two electrical points, which carry the signal, with respect to a third point. That is the noise should be common to both the input terminals. To clarify, suppose a small signal, of the order of a few mV, appears between two electrical terminals. Now if an external noise with respect to a third terminal appears on both the above two terminals with equal magnitude and phase then it is possible to eliminate the noise.

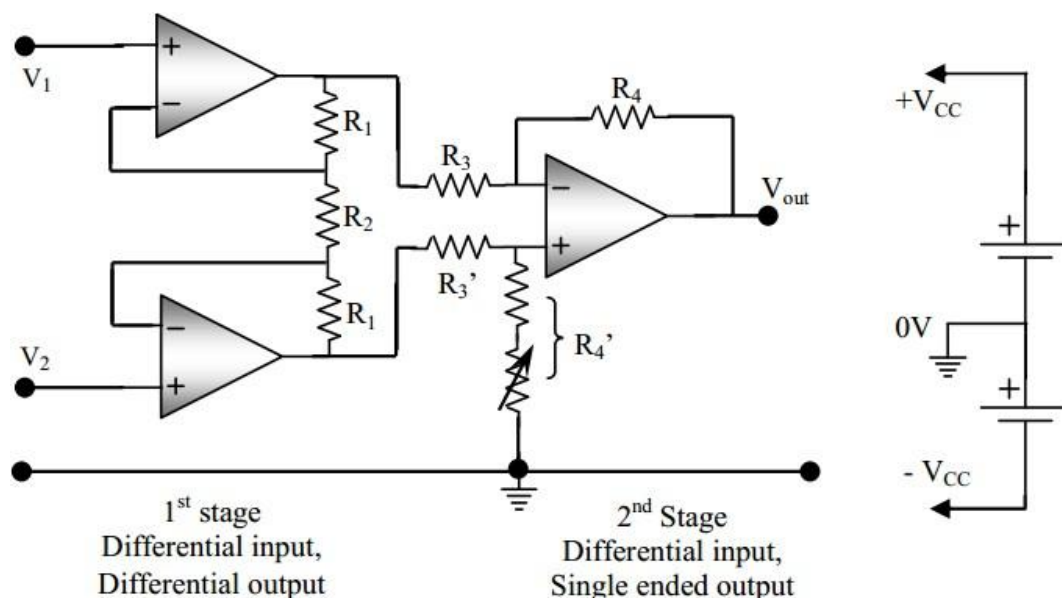


Figure 01: Instrumentation Amplifier

Instrumentation Amplifier:

Figure 01 shows the basic circuitry of an instrumentation amplifier consisting of 3 operational amplifiers (Op-amp). This is the most popular form of an instrumentation amplifier which has two inputs V_1 and V_2 , and one output V_{out} , all referenced to ground or common (0V). Here we

will perform the analyses based on ideal assumptions for the op-amps, and will also assume that A_{CM} is zero, initially. Here,

$$V_{out} \propto (V_2 - V_1)$$

i.e. it is a differential amplifier where the output is proportional to the difference of two inputs. It has two stages as indicated in **Figure 01**. The first stage has a differential input and also has a differential output (double ended input, V_1 and V_2 and double ended output, V_1' and V_2'), while the 2nd stage has a differential input (double ended input, V_1' and V_2') and a single ended output V_{out} .

In the circuit shown, Resistor R_4' has been shown to consist of a fixed resistor in series with a variable resistor. This is required to adjust the Common Mode Rejection Ratio (CMRR).

Common Mode Gain:

Common mode voltage (V_{CM}) is the voltage that is common to the two inputs of the instrumentation amplifier.

Common mode voltage gain (A_{CM}) is the ratio of the output voltage and input voltage when a common mode voltage is applied. Theoretically, the common mode voltage is expected to be zero,

$$\text{i.e. } V_2 - V_1 = 0$$

If the common mode voltage is not zero, then there will be a contribution of this voltage in the output.

$$V_{OUT} = A_{CM} V_{CM}$$

From here, we can measure A_{CM} .

Common Mode Rejection Ratio (CMRR):

CMRR should be as large as possible. For an ECG amplifier, it should be > 60 dB, and for EEG it should be > 100 dB. In the 2nd stage of the amplifier two ratios, R_4 / R_3 and R_4' / R_3' , associated with the two inputs of the differential amplifier are assumed to be equal. This is essential to get a high value of CMRR. However, in practice it is almost impossible to have two exactly equal fixed resistors, so this condition is achieved manually, through experiment. Therefore, R_4' is configured as a variable resistor in series with a fixed resistor, and a maximum CMRR is achieved through experimental adjustment of the variable resistor

Equipment: (Specify the model and quantities)

- Op-Amps
- Resistors
- Variable resistors
- DC power source
- Signal Generator
- Oscilloscope

Differential Gain vs Frequency

Differential Gain in dB vs Frequency

Common Mode Gain vs Frequency

Common mode Gain in dB vs Frequency

CMRR vs Frequency

Results:

The results of this experiment are the values of differential and common mode gains and the CMRR that you obtained at different frequencies. Present the results in graphical form in the main text and place the raw tables in an Appendix at the end.

Although indicated above, to summarize, you need the following plots from your measured data.

- Ad (linear scale) against frequency (log scale).
- Ad (dB scale) against frequency (log scale).
- CMRR (in linear scale) with frequency (log scale).
- CMRR (in dB scale) with frequency (log scale).

Discussion:

Discuss the problems, findings, shortcomings, outcome discrepancies of the experiment with theoretical explanation.